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Introduction

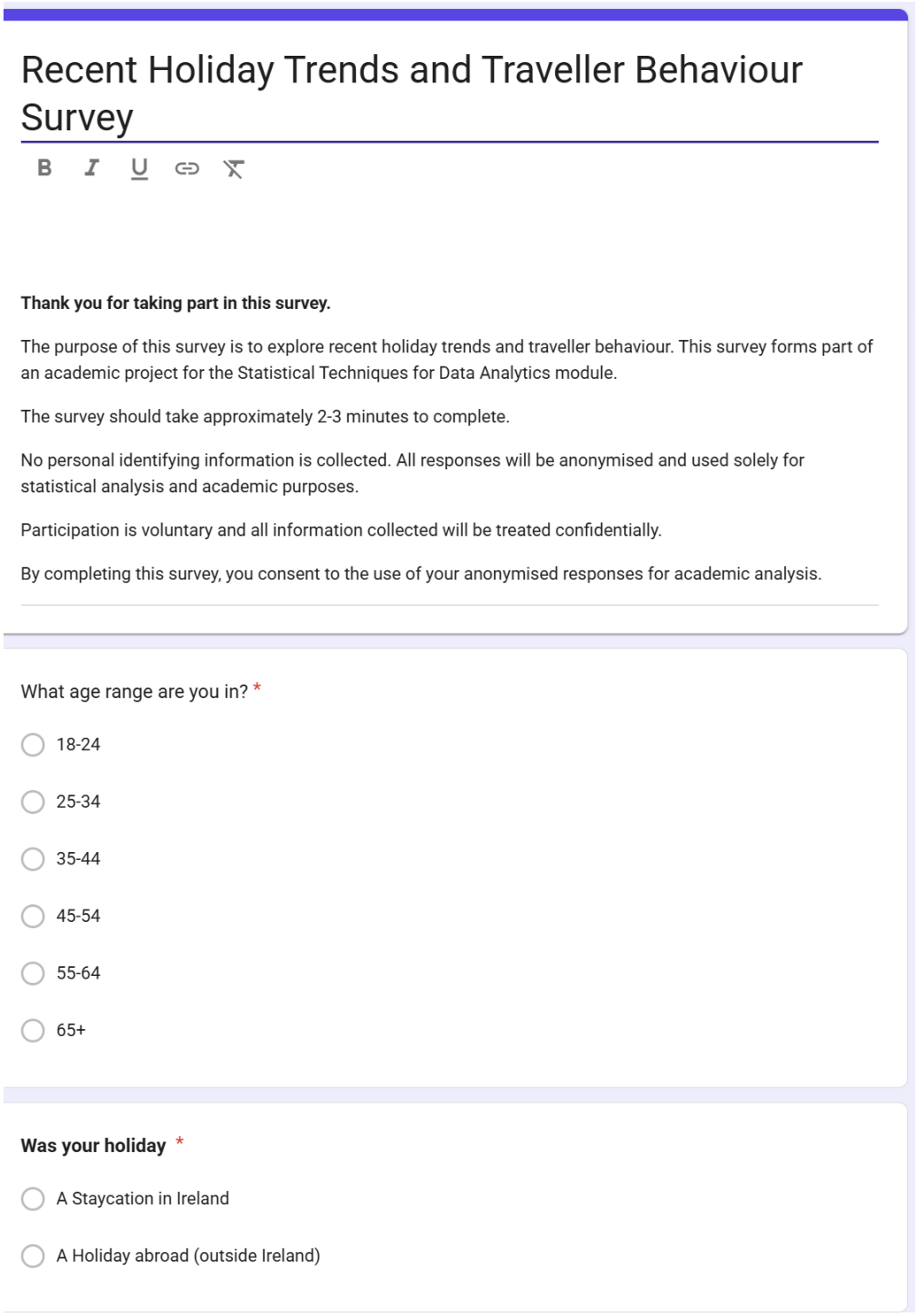
The purpose of this study was to investigate holiday trends and traveller behaviour. As the summer period was just ending, I thought it would be a valid time to complete a survey. With today's cost of living issues, I wanted to gauge if people are still willing to take summer holidays and get an understanding of the overall costs involved.

As part of the survey, I wanted to find out if people are still going overseas or preferring to stay in Ireland. Other areas of the survey were to look at the age range of respondents to get an understanding of how they book holidays and how much on average the holiday costs (overall and accommodation costs).

The survey was designed with Google Forms and called:

Recent Holiday Trends and Traveller Behaviour Survey

A link to the survey was issued to friends, family and fellow students on my Phone contacts. Everyone was notified that this survey was for academic purposes only and all responses will be anonymised.



Recent Holiday Trends and Traveller Behaviour Survey

Thank you for taking part in this survey.

The purpose of this survey is to explore recent holiday trends and traveller behaviour. This survey forms part of an academic project for the Statistical Techniques for Data Analytics module.

The survey should take approximately 2-3 minutes to complete.

No personal identifying information is collected. All responses will be anonymised and used solely for statistical analysis and academic purposes.

Participation is voluntary and all information collected will be treated confidentially.

By completing this survey, you consent to the use of your anonymised responses for academic analysis.

What age range are you in? *

- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65+

Was your holiday *

- A Staycation in Ireland
- A Holiday abroad (outside Ireland)

Figure 1 shows a snapshot of the Google Form

Data Preparation

The questions were designed in a way to give us data on:

- Categorical Variables
- Numerical Variables
- Likert Scale
- Boolean Variables

Data Preparation Actions

Once completed, I received 71 responses on Google Forms from my survey. This survey was downloaded as a CSV file and named:

D**ata_CA1.csv**

The next stage was to clean the data, such as:

- Trim any leading spaces from the form
- Check for null values
- Check shape
- Check the information on loaded data
- Rename columns to make it more readable

No missing values were discovered on the dataset and was now ready for analysis.

Trim any spaces from the headings

[47]:
```
df_Holidays.columns = df_Holidays.columns.str.strip()
```

Missing Value Analysis

[48]:
```
df_Holidays.isnull().sum()
```

[48]:
```
Timestamp
0
What age range are you in?
0
Was your holiday
0
What was the main purpose of your Holiday
0
How did you book your holiday
0
How recent was this holiday
0
How would you describe your holiday type
0
How many people were in your travel group?
0
Approximately how much did your accommodation cost per room per night (€)?
0
How many nights did your holiday last?
0
Overall Holiday Cost - Approximate total spend for all travellers (€), including  cost of holiday + daily spending.
0
How satisfied were you with your holiday overall?
0
Would You Take This Type of Holiday Again?
0
Did You Travel by Air?
0
How many holidays have you taken in the last 12 months?
0
dtype: int64
```

No missing values were identified in the dataset. Therefore no records required removal

No missing values were identified in the dataset. Therefore no records required removal

[49]:
```
df_Holidays.shape
```

[49]:
```
(71, 15)
```

[50]:
```
df_Holidays.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 71 entries, 0 to 70
Data columns (total 15 columns):
 #   Column                                                                                                  Non-Null Count  Dtype 
---  ------                                                                                                  --------------  ----- 
 0   Timestamp                                                                                               71 non-null     object
 1   What age range are you in?                                                                              71 non-null     object
 2   Was your holiday                                                                                        71 non-null     object
 3   What was the main purpose of your Holiday                                                               71 non-null     object
 4   How did you book your holiday                                                                           71 non-null     object
 5   How recent was this holiday                                                                             71 non-null     object
 6   How would you describe your holiday type                                                                71 non-null     object
 7   How many people were in your travel group?                                                              71 non-null     int64 
 8   Approximately how much did your accommodation cost per room per night (€)?                              71 non-null     int64 
 9   How many nights did your holiday last?                                                                  71 non-null     int64 
 10  Overall Holiday Cost - Approximate total spend for all travellers (€), including  cost of holiday + daily spending.  71 non-null     int64 
 11  How satisfied were you with your holiday overall?                                                       71 non-null     int64 
 12  Would You Take This Type of Holiday Again?                                                              71 non-null     object
 13  Did You Travel by Air?                                                                                  71 non-null     object
 14  How many holidays have you taken in the last 12 months?                                                 71 non-null     int64 
dtypes: int64(6), object(9)
memory usage: 8.4+ KB
```

Rename columns to make it more readable

[51]:
```
df_Holidays = df_Holidays.rename(columns={
    "Timestamp": "Timestamp",
    "What age range are you in?": "Age_Range",
    "Was your holiday": "Holiday_Location",
    "What was the main purpose of your Holiday": "Holiday_Purpose",
    "How did you book your holiday": "Booking_Method",
    "How recent was this holiday": "Holiday_Recency",
    "How would you describe your holiday type": "Holiday_Type",
    "How many people were in your travel group?": "Group_Size",
    "Approximately how much did your accommodation cost per room per night (€)?": "Accommodation_Cost",
    "How many nights did your holiday last?": "Holiday_Length",
    "Overall Holiday Cost - Approximate total spend for all travellers (€), including  cost of holiday + daily spending.": "Total_Cost",
    "How satisfied were you with your holiday overall?": "Satisfaction",
    "Would You Take This Type of Holiday Again?": "Holiday_Again",
    "Did You Travel by Air?": "Travel_By_Air",
    "How many holidays have you taken in the last 12 months?": "Holidays_Last_12_Months"
    }
    )
```

Descriptive Analysis

Descriptive statistics were carried out on the survey dataset to identify patterns from respondents.

Descriptive Statistics

```
df_Holidays.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Group_Size	71.0	3.253521	1.754328	1.0	2.0	3.0	4.0	10.0
Accommodation_Cost	71.0	198.295775	478.726686	1.0	80.0	115.0	150.0	4000.0
Holiday_Length	71.0	7.267606	5.022685	1.0	4.0	6.0	7.5	28.0
Total_Cost	71.0	2285.915493	1611.923941	200.0	850.0	2000.0	3000.0	6000.0
Satisfaction	71.0	4.478873	0.673141	3.0	4.0	5.0	5.0	5.0
Holidays_Last_12_Months	71.0	3.000000	3.171300	1.0	2.0	2.0	3.0	22.0

Median Analysis

```
df_Holidays.median(numeric_only=True)
```

```
Group_Size                  3.0
Accommodation_Cost        115.0
Holiday_Length              6.0
Total_Cost               2000.0
Satisfaction                5.0
Holidays_Last_12_Months     2.0
dtype: float64
```

The average Accommodation cost (per night) was €198.29. However, the median analysis shows the accommodation cost of €115.00. This would suggest that the distribution is positively skewed due to a few unusually high accommodation costs. This is further supported with a max value of €4,000.

Skewness analysis was presented to understand the shape of the distributions. As expected, Accommodation Cost displayed a high positive skewness of 7.43, confirming a few extreme outliers.

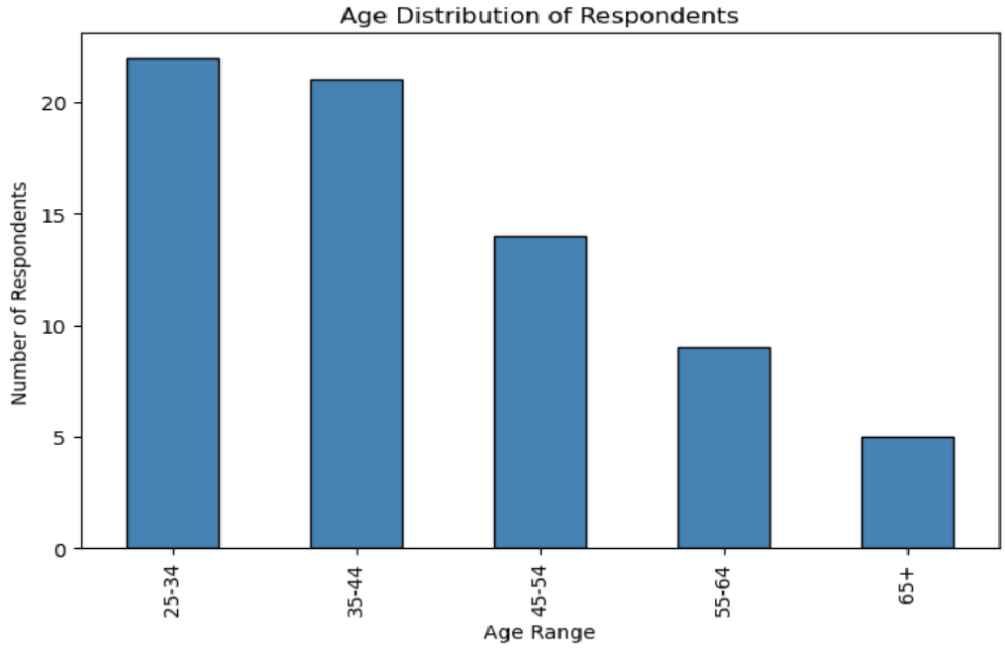
The mean Holiday length showed as 7.26 days, whilst it showed a positive skewness of 2.55, again resulting an outlier, with a max value of 28 days, influencing the overall distribution.

Skewness Analysis

```
df_Holidays.skew(numeric_only=True)
```

```
Group_Size                 1.543669
Accommodation_Cost         7.432945
Holiday_Length             2.559220
Total_Cost                 0.743280
Satisfaction              -0.933230
Holidays_Last_12_Months    4.681742
dtype: float64
```

For a categorical Analysis, an age bar chart that was created shows that respondents were reasonably evenly distributed across the younger and middle-age categories. The 25–34 and 35–44 age groups represented the largest proportion of respondents, while the 65+ category contained the fewest responses.



Online bookings were the most popular method, with approximately 78% of all bookings, as opposed to package holidays.

Show Booking methods as a Percentage

```
8]: df_Holidays['Booking_Method'].value_counts(normalize=True) * 100
```

```
8]: Booking_Method
    Independently Booked online    78.873239
    A package holiday              21.126761
    Name: proportion, dtype: float64
```

Beach holidays abroad were the most popular holiday type. This is not surprising as the survey took place immediately after the summer holiday season.

```
df_Holidays['Holiday_Purpose'].value_counts()
```

```
Holiday_Purpose
Beach Holiday abroad                              37
City Break                                        11
To visit family / Friends                         11
Other                                              7
To attend an event (concert, Football match etc)   5
Name: count, dtype: int64
```

Overall, the descriptive statistics provided a better understanding of holiday spending patterns, satisfaction ratings and holiday types. However, we can see a few outliers in the dataset that warranted further investigation.

Analysis & Insights

Booking Methods by Age Range

See below: pivot table was created to give an analysis of Booking methods by age range.

Run a pivot table to show Booking Methods broken down by Age Range

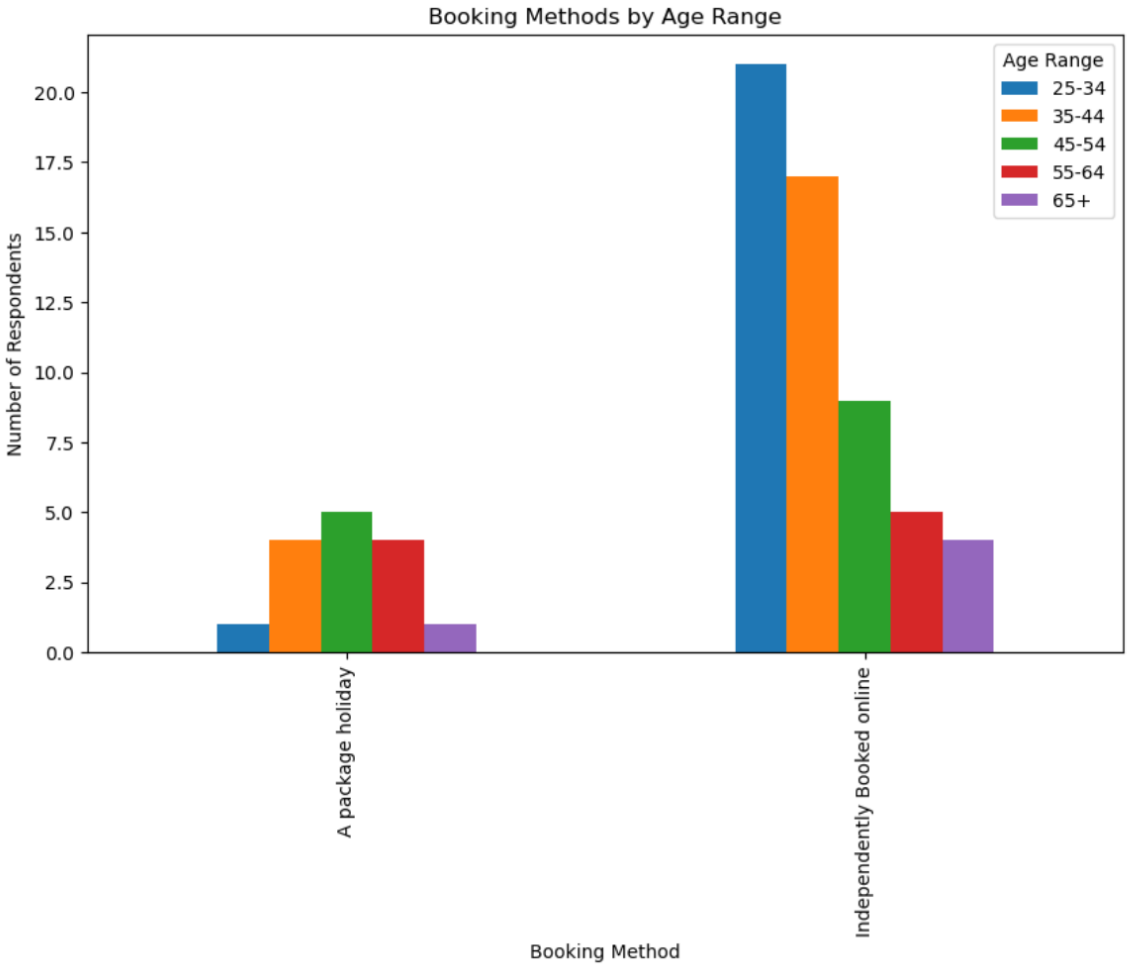
[59]:
```
# create a pivot table
pivot_bookings = pd.pivot_table (
    df_Holidays,
    values="Group_Size",
    index="Booking_Method",
    columns="Age_Range",
    aggfunc="count",
    margins=True,
    margins_name="Total"
)
pivot_bookings
```

[59]:

Age_Range	25-34	35-44	45-54	55-64	65+	Total
Booking_Method						
A package holiday	1	4	5	4	1	15
Independently Booked online	21	17	9	5	4	56
Total	22	21	14	9	5	71

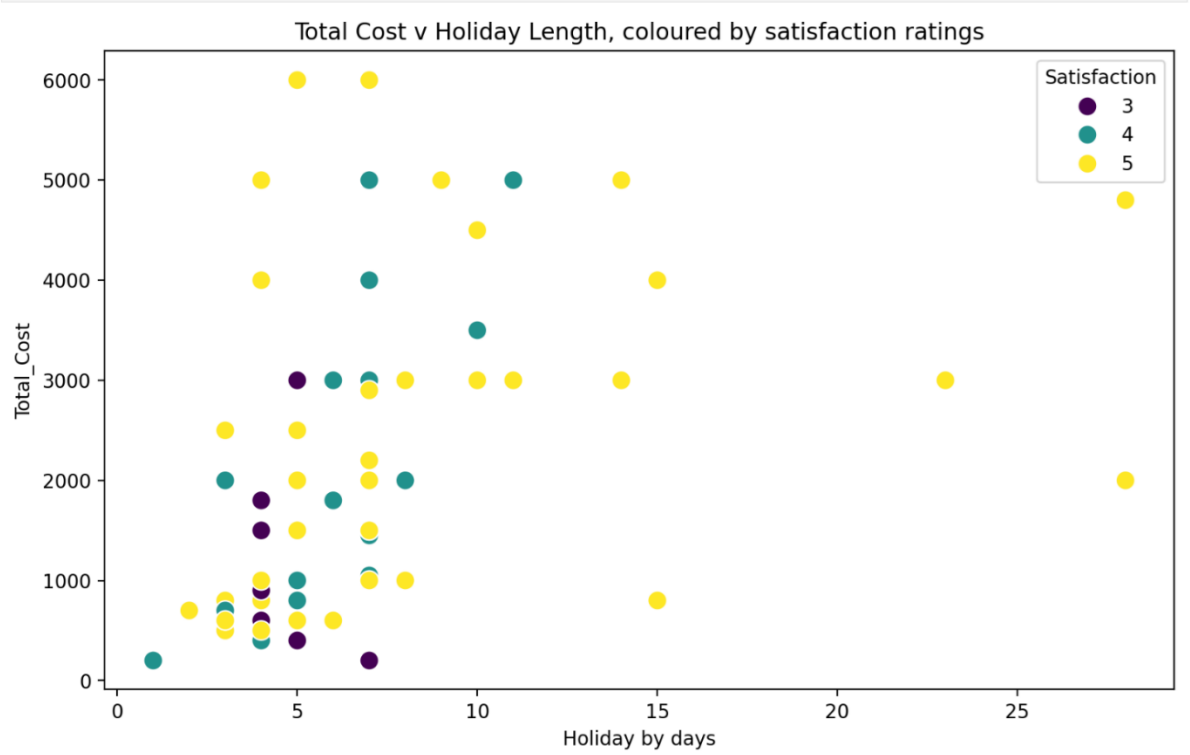
By breaking down Booking Methods by age range, we can see that online bookings remained the preferred choice of booking across all age categories, although package holidays were more common among the respondents aged 45-54.

A bar chart was used to further visualise the results



Relationship to Holiday Duration, Total Expenditure & overall Satisfaction

In this analysis, a scatterplot was used to visualise this multivariate analysis

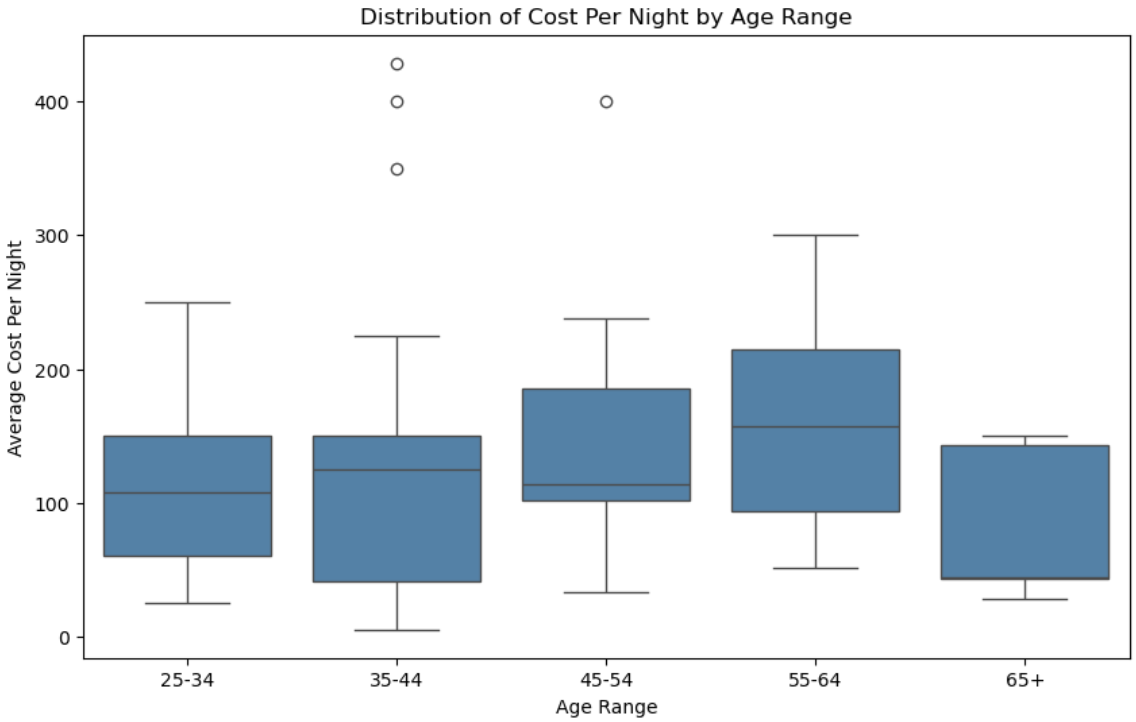


Scatterplot Analysis

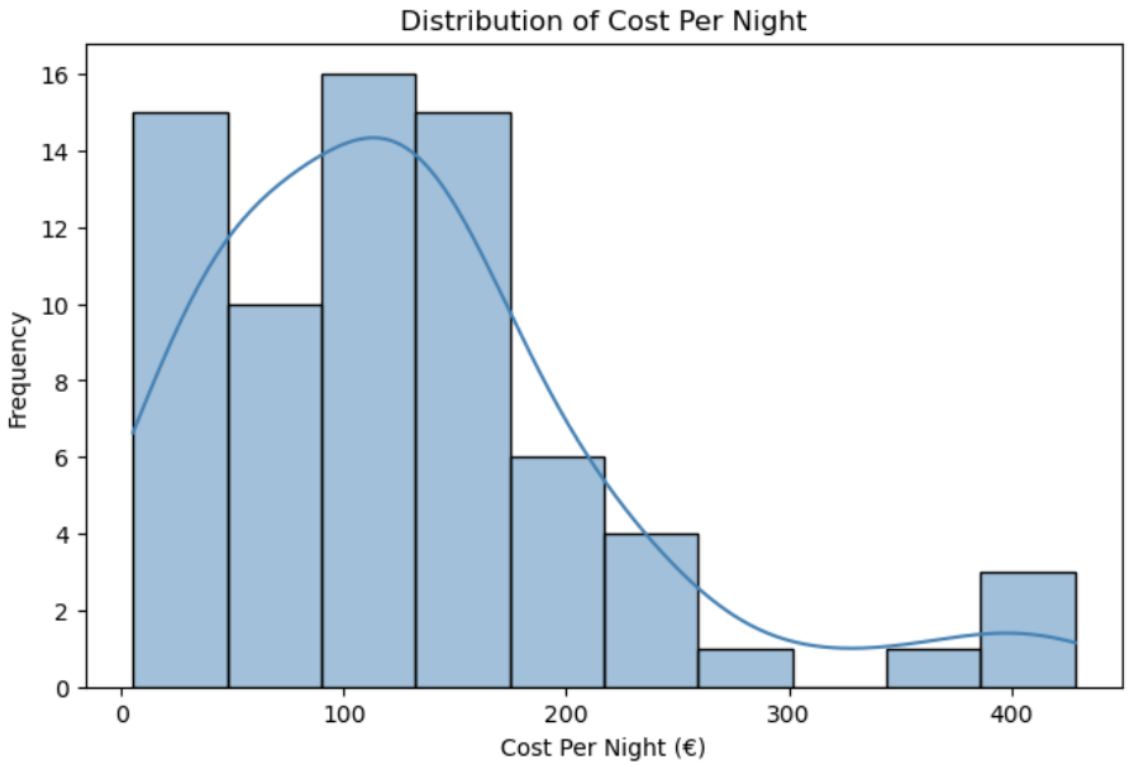
The scatterplot shows a positive relationship between duration of holiday and total cost. However, a higher satisfaction rating can be observed across both low-cost and high-cost holidays.

Correlation and Outlier Analysis

A streamlined data frame was created along with a new derived variable called “Cost_Per_Night”. This was created by dividing Total_Cost by Holiday_Length and then dividing by Group_Size.

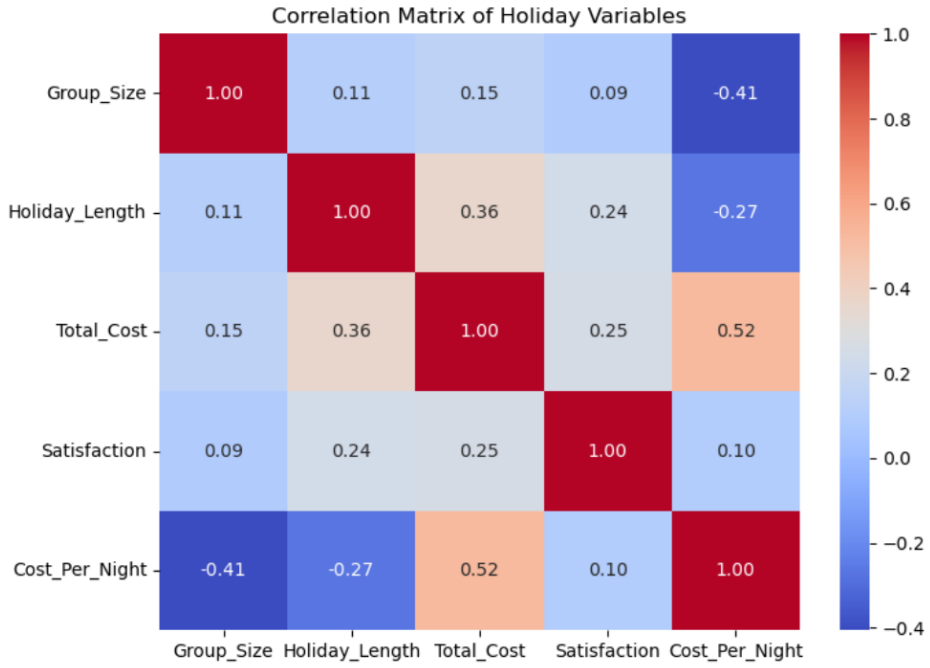


The presence of outliers helps explain why some age groups have noticeably higher mean values in the previous bar chart



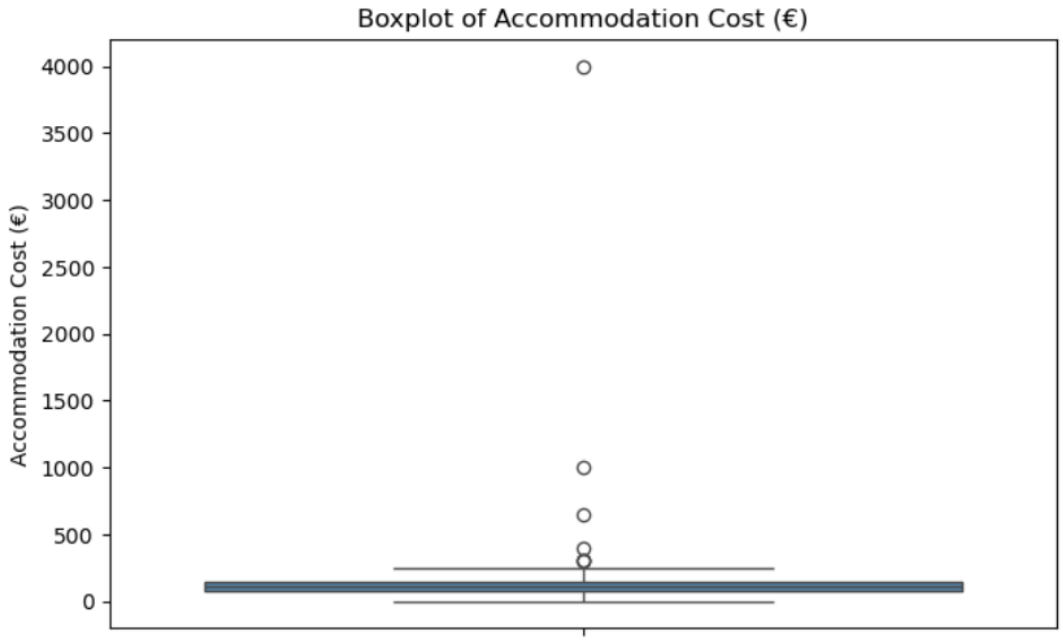
The histogram of Cost_Per_Night demonstrates a positively skewed distribution. Most respondents recorded a cost per night below €300, whilst a smaller number of respondents reported a significantly higher spending. The presence of these higher values creates a longer tail on the right-hand side of the distribution, indicating potential outliers.

Correlation Analysis

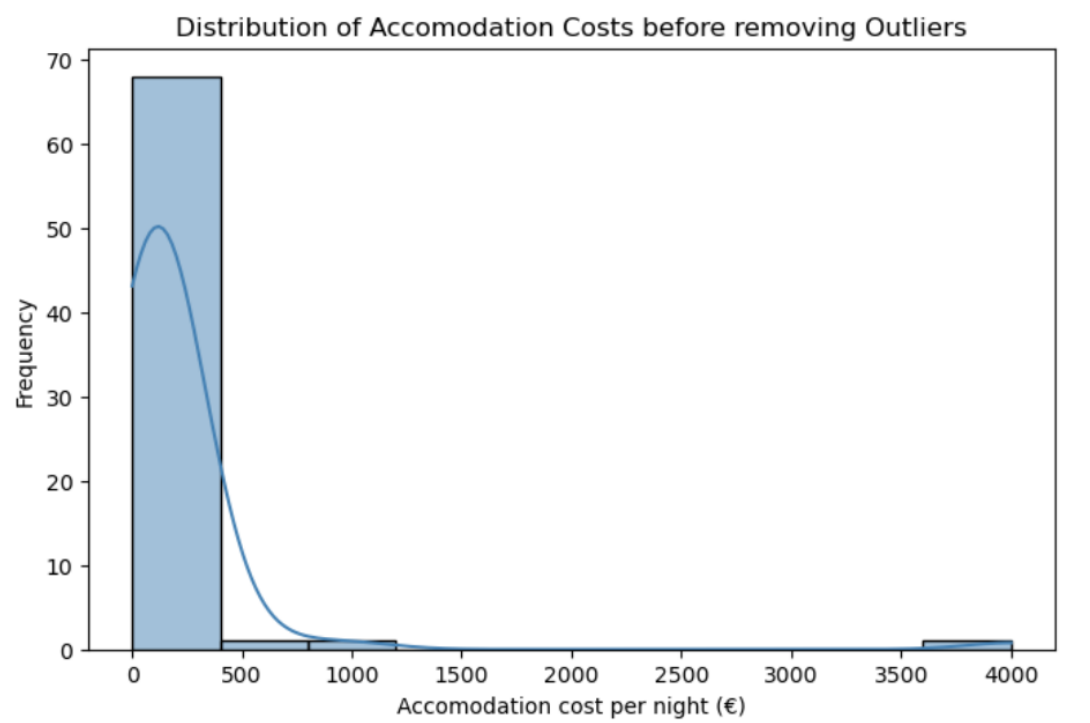


The strongest positive correlation was between Total_Cost and Cost_Per_Night (r=0.52). This indicates that respondents who spent more overall also tended to spend more on a per-night basis, which is to be expected.

Outlier Analysis using Accomodation Costs



The Boxplot clearly shows the presence of several extreme accommodation costs.



Drop Outliers in Accomodation Costs using the IQR Method

```
Q1 = df_Accom["Accommodation_Cost"].quantile(0.25)
Q3 = df_Accom["Accommodation_Cost"].quantile(0.75)
IQR = Q3 - Q1
print("Q1= " , Q1)
print("Q3= " , Q3)
print("IQR= " , IQR)
```

```
Q1=  80.0
Q3=  150.0
IQR=  70.0
```

```
lower_bound = Q1 - (1.5 * IQR)
upper_bound = Q3 + (1.5 * IQR)

print("lower_bound =  " , lower_bound)
print("upper_bound =  " , upper_bound)
```

```
lower_bound =   -25.0
upper_bound =   255.0
```

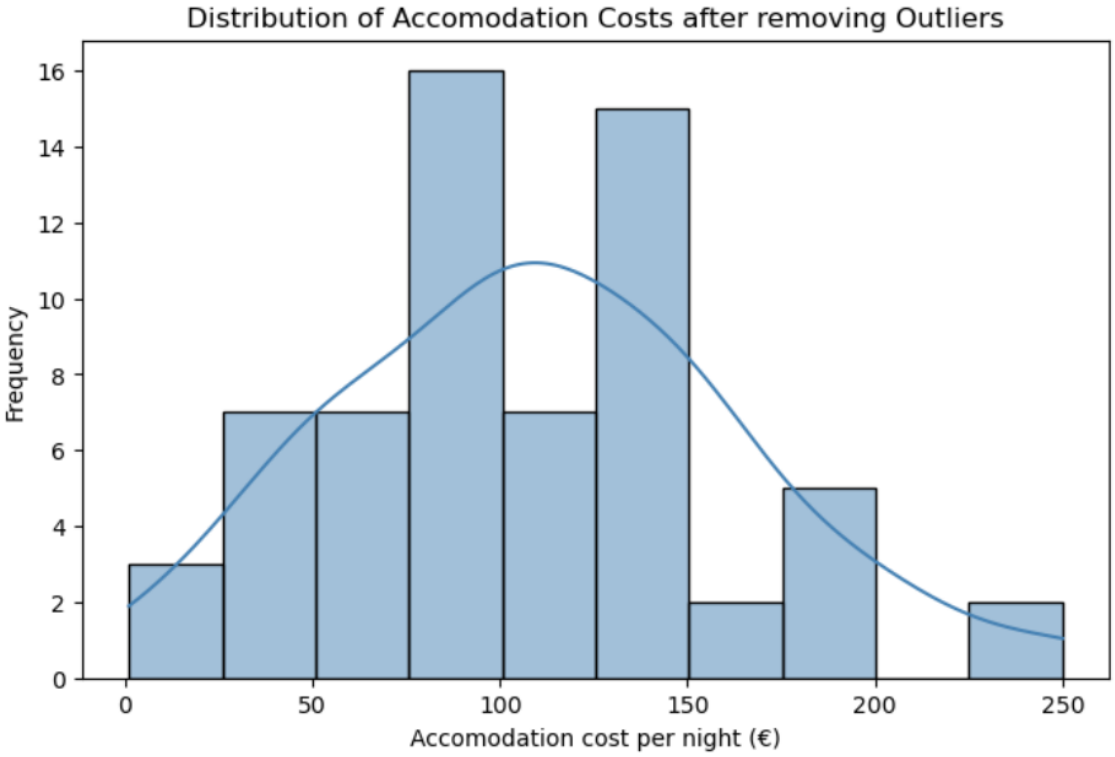
remove the lower and upper bounds

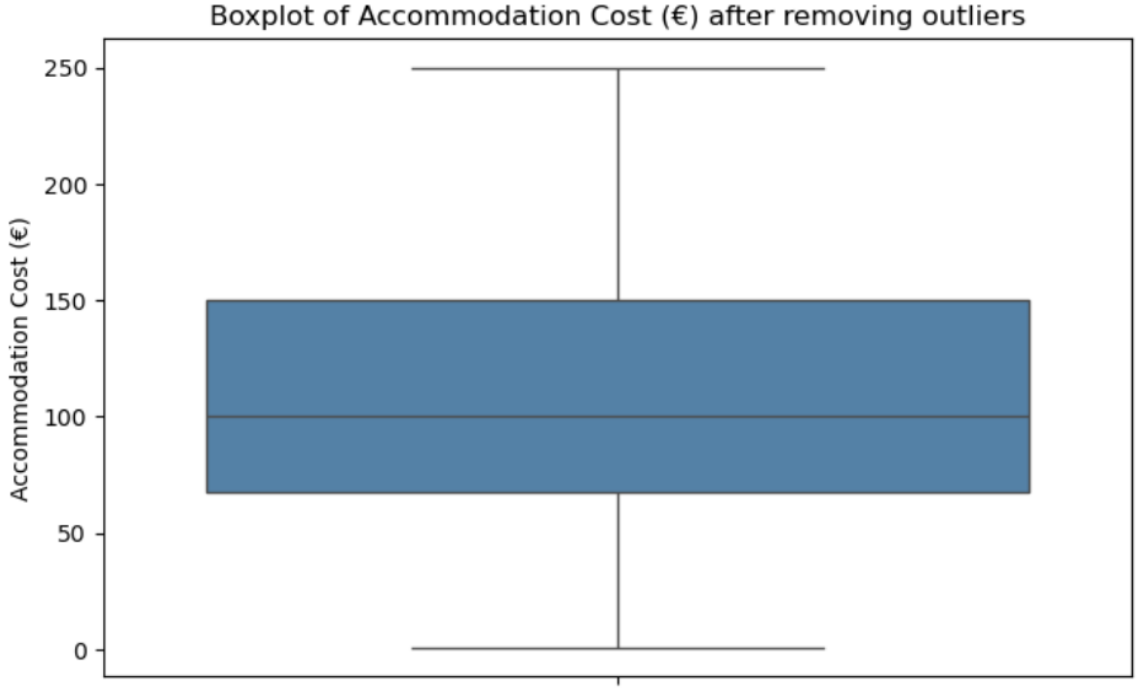
```
df_Accom = df_Accom[
    (df_Accom["Accommodation_Cost"] >= lower_bound) &
     (df_Accom["Accommodation_Cost"] <= upper_bound)
      ]
```

```
df_Accom["Accommodation_Cost"].describe()
```

```
count     64.000000
mean     111.390625
std       54.590719
min        1.000000
25%       67.500000
50%      100.000000
75%      150.000000
max      250.000000
Name: Accommodation_Cost, dtype: float64
```

The mean accommodation costs of €111.39 are recorded after removal of outliers. The original dataset has a mean of €198.28 before removal of outliers. This further indicates that a small number of observations were increasing the average cost.


Distribution of Accomodation Costs after removing Outliers
Frequency
Accomodation cost per night (€)
0
2
4
6
8
10
12
14
16
0
50
100
150
200
250


Boxplot of Accommodation Cost (€) after removing outliers
Accommodation Cost (€)
0
50
100
150
200
250

Using the IQR Method to identify outliers with the Accommodation Cost variable

- Q1 was 80
- Q3 was 150
- IQR was 70

By applying the standard 1.5 x IQR the upper bound was €255 and the lower bound was €-25 were calculated. Therefore, only Rows exceeding the upper bounds were classified as outliers and removed. This resulted in seven observations classified as outliers and removed, with a new dataset of 64 remaining.

Conclusion and Discussion

The analysis provided valuable insights into the Holiday Trends survey, showing us that online bookings were the preferred method across all age groups, while longer holidays resulted in higher expenditure. Satisfaction levels were high across the dataset, regardless of expenditure.

The impact of outliers on the Accommodation Cost variable was a main finding. Extreme values resulted in a positively skewed distribution. Once outliers were removed, using the IQR method, the distribution became more representative of typical accommodation costs.

Using Python and its libraries, a raw survey dataset and transform it into meaningful insights, through statistical analysis and data visualisation techniques, enabling trends and unusual observations to be identified.